2. Introduction and Background

Week 3 focused on three menu-driven Python applications: Bank Management System, Hotel Booking System, and Library Management System. All use in-memory data structures, a common boxed terminal interface, and menu-driven control flow. This report presents each program separately, followed by a combined timeline, tools, and references

3. Program 1: Bank Management System

3.1 Introduction and Background

A Bank Management System simplifies banking operations such as account creation, balance enquiry, deposits, and withdrawals. This project develops a console-based Python application using in-memory data structures, featuring unique account generation, PIN authentication, and a boxed terminal interface.

3.2 Objectives

- Develop a menu-driven bank management system.
- Generate unique account numbers.
- Implement PIN-based authentication.
- Support deposit, withdrawal, and balance enquiry.
- Build a consistent terminal interface.

3.3 Problem Statement / Driving Question

How can basic banking operations be implemented securely in a menu-driven console application using Python without an external database?

3.4 Methodology

- **Program Design:** Created a boxed terminal interface using reusable helper functions and an ASCII-art banner.
- **Data Structures:** Used a USERS dictionary to store account details and an ACCOUNTS list to maintain unique account numbers.
- **Core Banking Functions:** Implemented account generation, account creation, authentication, deposit, and withdrawal with balance validation.
- **Menu-Driven Control Flow:** Built a continuous menu with Create Account, Check Balance, Withdraw, Deposit, and Exit options, along with input validation and error handling.

3.5 Expected Outcomes / Deliverables

- A functional bank management system.
- Unique account number generation.
- Secure PIN-based authentication.
- Input validation and error handling.
- A user-friendly boxed terminal interface